

Sparse SLAM

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1 Introduction

The current goal is to improve SLAM with one of the following techniques:

- **Graph Sparsity Parameters.**
- **Positivity Theorems.**

2 Related Work

This section, for now, contains the related paper that we have founds for each approach.

2.1 Graph Sparsity

Foundations of spatial perception for robotics: Hierarchical representations and real-time systems [1].

2.2 Positivity Theorems

3 Preliminaries

SLAM.

Pose-Graph SLAM.

4 Algorithm

5 Experiments

References

- [1] Nathan Hughes, Yun Chang, Siyi Hu, Rajat Talak, Rumaia Abdulhai, Jared Strader, and Luca Carlone. Foundations of spatial perception for robotics:

Hierarchical representations and real-time systems. *The International Journal of Robotics Research*, 43(10):1457–1505, 2024.